HW10

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Problem 1

(a)

Let $P = \text{Unif}[0,1],\ Q = \text{Unif}[0,2]$. Then on $[0,1]$, both densities are positive, so

$$D_{\text{KL}}(P\|Q) = \int_0^1 \log\left(\frac{1}{1/2}\right) dx = \int_0^1 \log 2 \, dx = \log 2 < \infty.$$

However, on $(1,2]$ we have $Q(x) > 0$ but $P(x) = 0$. Hence

$$D_{\text{KL}}(Q\|P) = \int_0^2 Q(x) \log\left(\frac{Q(x)}{P(x)}\right) dx = \infty,$$

since the integrand is infinite on a region of positive measure.Thus $D_{\text{KL}}(P\|Q) < \infty$ while $D_{\text{KL}}(Q\|P) = \infty$.

(b)

Forward KL is $D_{\text{KL}}(P\|Q)$, which heavily penalizes regions where $P(x)$ is large but $Q(x)$ is small. Reverse KL is $D_{\text{KL}}(Q\|P)$, which penalizes placing mass where $P(x)$ is small.

In B, Q concentrates on a single mode of P and ignores the other. This makes Q small where P is large, which would cause forward KL to be large, but reverse KL would be moderate since Q does not assign probability where P is near zero. Thus B corresponds to minimizing reverse KL. In figure A, Q spreads mass between the two modes to avoid being too small where P is large. This is characteristic of minimizing forward KL. So, A would be forward KL and B reverse KL.

Problem 2

(a)

To sample $Z \sim \mathcal{N}(\mu_z, \sigma_z^2)$ using $V \sim \mathcal{N}(0,1)$, we write $Z = \sigma_z V + \mu_z$. So, in our diagram, we would multiply V by σ_z, add μ_z, and produce Z.

(b)

The objective is

$$L = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}_{p_{\theta_e}(z|x_n)}[-\log q_{\theta_d}(y_n|z)] + \beta D_{\text{KL}}(p_{\theta_e}(Z|x_n)\|r(Z)).$$

The loss depends on both θ_e (through z) and θ_d.The KL term depends only on θ_e. Thus,

$$\text{Encoder } (\theta_e) : \text{task loss and KL term}$$

$$\text{Decoder } (\theta_d) : \text{task loss only}$$

(c)

As β increases, regularization first improves generalization by preventing overfitting. After a certain point, the KL term dominates and forces Z to ignore label-relevant information, worsening validation performance. Thus the curve that decreases and then increases (U-shaped) corresponds to the correct trend, which is figure (b).

(d)

(i) When β is very large, the latent codes collapse to a standard Gaussian centered at zero. When β is very small, the latent codes spread to fit the labels. From the plots: $(a)\ \beta = 10^{-3},\ (b)\ \beta = 10^{-6},\ (c)\ \beta = 10^{0}$.

(ii) A good latent space should separate classes clearly while remaining structured. Figure (a) shows distinct clusters without collapse, which is the best representation.

Problem 4

(a)

First, we can write the following:

$$L^{\text{minimax}}_{\text{generator}}(\theta;\phi) = \mathbb{E}_{z\sim\mathcal{N}(0,I)}\big[\log(1-\sigma(h_\phi(G_\theta(z))))\big].$$

Using $\sigma'(x) = \sigma(x)(1-\sigma(x))$, we compute the gradient with respect to θ, so

$$\nabla_\theta L^{\text{minimax}}_{\text{generator}} = \mathbb{E}_z\left[\frac{-\sigma'(h_\phi(G_\theta(z)))}{1-\sigma(h_\phi(G_\theta(z)))}\nabla_\theta h_\phi(G_\theta(z))\right].$$

Substituting $\sigma'(x)$ and simplifying,

$$\nabla_\theta L^{\text{minimax}}_{\text{generator}} = -\mathbb{E}_z\big[\sigma(h_\phi(G_\theta(z)))\nabla_\theta h_\phi(G_\theta(z))\big].$$

Since $D_\phi(G_\theta(z)) = \sigma(h_\phi(G_\theta(z)))$, we see that when the discriminator confidently rejects fake samples,

$$D_\phi(G_\theta(z)) \approx 0 \quad\Longrightarrow\quad \nabla_\theta L^{\text{minimax}}_{\text{generator}} \approx 0.$$

Thus the generator gradient vanishes, slowing or stalling learning when the discriminator is strong.

(b)

Optimal Discriminator. For fixed generator parameters θ, the discriminator objective is

$$L_{\text{disc}}(\phi;\theta) = -\mathbb{E}_{x\sim p_{\text{data}}}[\log D_\phi(x)] - \mathbb{E}_{x\sim p_\theta}[\log(1-D_\phi(x))].$$

Writing the expectations as integrals,

$$L_{\text{disc}} = -\int p_{\text{data}}(x)\log D_\phi(x)\,dx - \int p_\theta(x)\log(1-D_\phi(x))\,dx.$$

Since the integrand depends on x only through $D_\phi(x)$, the minimization can be performed pointwise. For each fixed x, define

$$f(D) = -p_{\text{data}}(x)\log D - p_\theta(x)\log(1-D), \quad D\in(0,1).$$

Differentiate with respect to D:

$$f'(D) = -\frac{p_{\text{data}}(x)}{D} + \frac{p_\theta(x)}{1-D}.$$

Setting $f'(D) = 0$ gives

$$\frac{p_\theta(x)}{1-D} = \frac{p_{\text{data}}(x)}{D},$$

$$p_\theta(x)D = p_{\text{data}}(x)(1-D),$$
$$D\big(p_\theta(x) + p_{\text{data}}(x)\big) = p_{\text{data}}(x),$$
$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)}.$$

To verify this is a minimum, compute

$$f''(D) = \frac{p_{\text{data}}(x)}{D^2} + \frac{p_\theta(x)}{(1-D)^2} > 0,$$

so f is strictly convex and the critical point is the unique minimizer. Therefore, the optimal discriminator is

$$D^*_\phi(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)}.$$

Value of the Minimax Objective. The minimax objective is

$$V(G, D) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{x\sim p_\theta}[\log(1 - D(x))].$$

Using the optimal discriminator,

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)},$$

$$V(G, D^*) = \mathbb{E}_{p_{\text{data}}}\left[\log \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)}\right] + \mathbb{E}_{p_\theta}\left[\log \frac{p_\theta(x)}{p_{\text{data}}(x) + p_\theta(x)}\right].$$

Rewrite each logarithm by factoring out a constant,

$$= -\log 4 + \mathbb{E}_{p_{\text{data}}}\left[\log \frac{2p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)}\right] + \mathbb{E}_{p_\theta}\left[\log \frac{2p_\theta(x)}{p_{\text{data}}(x) + p_\theta(x)}\right].$$

The expression is minimized when the two distributions coincide, so

$$p_\theta(x) = p_{\text{data}}(x),$$

in which case the objective achieves its minimum value

$$V(G, D^*) = -\log 4.$$

Problem 5

(a)

The forward diffusion process is defined by $q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I)$. Let $\alpha_t = \prod_{i=1}^{t}(1-\beta_i)$. Using repeated substitution and the Gaussian reparameterization trick, we can write $x_t = \sqrt{\alpha_t}\, x_0 + \sqrt{1-\alpha_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$. Therefore, the marginal distribution is as following: $q(x_t \mid x_0) = \mathcal{N}(\sqrt{\alpha_t}\, x_0,\, (1-\alpha_t)I)$.

(b)

By Bayes' rule, $q(x_{t-1} \mid x_t, x_0) \propto q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)$. Both terms are Gaussian, so

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\, x_{t-1}, \beta_t I), \quad q(x_{t-1} \mid x_0) = \mathcal{N}(\sqrt{\alpha_{t-1}}\, x_0, (1-\alpha_{t-1})I).$$

The product of Gaussians gives another Gaussian, so $q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\mu_t(x_t, x_0),\, \hat{\beta}_t I)$, where

$$\hat{\beta}_t = \frac{1-\alpha_{t-1}}{1-\alpha_t}\, \beta_t, \qquad \mu_t(x_t, x_0) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\alpha_t} x_0.$$

Thus, the reverse conditional distribution is tractable and Gaussian.

Problem 6

(a)

For any positive function $g(\theta)$,

$$\nabla_\theta \log g(\theta) = \frac{\nabla_\theta g(\theta)}{g(\theta)}.$$

With $g(\theta) = p_\theta(x)$, we have

$$\nabla_\theta \log p_\theta(x) = \frac{\nabla_\theta p_\theta(x)}{p_\theta(x)}.$$

Multiplying both sides by $p_\theta(x)$,

$$p_\theta(x) \nabla_\theta \log p_\theta(x) = \nabla_\theta p_\theta(x),$$

as desired.

(b)

Let $\tau = (x, y)$. By definition,

$$J(w) = \mathbb{E}_{\tau \sim p_w}[L(\tau)] = \int p_w(\tau) L(\tau)\, d\tau.$$

Differentiating, we see

$$\nabla_w J(w) = \int \nabla_w\big(p_w(\tau) L(\tau)\big)\, d\tau.$$

The loss depends on w only through the sampling distribution, so $L(\tau)$ is treated as constant with respect to the gradient, and so

$$\nabla_w J(w) = \int L(\tau) \nabla_w p_w(\tau)\, d\tau,$$

$$\nabla_w p_w(\tau) = p_w(\tau) \nabla_w \log p_w(\tau),$$
$$\nabla_w J(w) = \int L(\tau)\, p_w(\tau) \nabla_w \log p_w(\tau)\, d\tau,$$
$$\nabla_w J(w) = \mathbb{E}_{\tau \sim p_w}\left[L(\tau) \nabla_w \log p_w(\tau)\right].$$

We now factor the joint distribution and take logs, so

$$p_w(\tau) = p_w(x, y) = p(x) p_w(y|x),$$

$$\log p_w(\tau) = \log p(x) + \log p_w(y|x).$$

As $p(x)$ is independent of w, we see

$$\nabla_w \log p_w(\tau) = \nabla_w \log p_w(y|x),$$

$$\nabla_w J(w) = \mathbb{E}_{(x,y)\sim p_w}\left[L(x, f(x))\,\nabla_w \log p_w(y|x)\right].$$

Now, replacing the expectation by a Monte-Carlo average,

$$\nabla_w J(w) \approx \frac{1}{N}\sum_{i=1}^{N} L(x_i, f(x_i))\,\nabla_w \log p_w(y_i|x_i).$$

Problem 8

(a)

From the update rule, $x_1 = \sqrt{\alpha_1}x_0 + \sqrt{1-\alpha_1}z_1, \quad z_1 \sim \mathcal{N}(0, I)$. A linear transformation of a standard Gaussian is Gaussian, hence $x_1|x_0 \sim \mathcal{N}(\sqrt{\alpha_1}x_0, (1-\alpha_1)I)$. Since $\bar{\alpha}_1 = \alpha_1$, the statement holds. Assume $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}z, \quad z \sim \mathcal{N}(0, I)$. Then, $x_{t+1} = \sqrt{\alpha_{t+1}}x_t + \sqrt{1-\alpha_{t+1}}z'$, with z' independent of z. Substituting the inductive hypothesis,

$$x_{t+1} = \sqrt{\alpha_{t+1}\bar{\alpha}_t}\,x_0 + \sqrt{\alpha_{t+1}(1-\bar{\alpha}_t)}\,z + \sqrt{1-\alpha_{t+1}}\,z'.$$

As $\bar{\alpha}_{t+1} = \alpha_{t+1}\bar{\alpha}_t$, the mean term becomes $\sqrt{\bar{\alpha}_{t+1}}x_0$. The remaining two terms are independent Gaussians. Their covariance adds, so

$$\alpha_{t+1}(1-\bar{\alpha}_t) + (1-\alpha_{t+1}) = 1-\bar{\alpha}_{t+1}.$$

Thus, they can be written as $\sqrt{1-\bar{\alpha}_{t+1}}z''$ for $z'' \sim \mathcal{N}(0, I)$, so we have as desired:

$$x_{t+1}|x_0 \sim \mathcal{N}(\sqrt{\bar{\alpha}_{t+1}}x_0, (1-\bar{\alpha}_{t+1})I).$$

(b)

All variables involved are jointly Gaussian, so the conditional distribution $x_{t-1}|x_t, x_0$ is Gaussian. So, we can write

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{\beta_t}z_t,$$
$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}x_0 + \sqrt{1-\bar{\alpha}_{t-1}}z.$$

Substituting the second into the first,

$$x_t = \sqrt{\alpha_t\bar{\alpha}_{t-1}}x_0 + \sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}z + \sqrt{\beta_t}z_t.$$

Thus (x_{t-1}, x_t) is an affine transformation of (z, z_t) and hence jointly Gaussian. For jointly Gaussian variables, the conditional mean is linear in (x_t, x_0), so

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0,$$

and the conditional variance is

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t.$$

Thus, the optimal predictor is $\tilde{\mu}_t(x_t, x_0)$.

(c)

Starting from $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} z$. If the network predicts $\epsilon_\theta(x_t, t)$ in place of z, then $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta(x_t, t)$. Solving equation for x_0, we see

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

Substituting into $\tilde{\mu}_t(x_t, x_0)$:

$$\mu_\theta(x_t, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \cdot \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta}{\sqrt{\bar{\alpha}_t}}.$$

We using $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$ to simplify the coefficients, so after canceling

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right).$$

Therefore,

$$\mu_\theta(x_t, t) = \frac{x_t - A\, \epsilon_\theta(x_t, t)}{B},$$

we have

$$A = \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}, \qquad B = \sqrt{\alpha_t}.$$